



Python Programming Foundations

Week 5 - Files, Errors, & Data Persistence
Day 2: "Structured Data & Basic Error Handling"

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Saved Data Has Shape

Yesterday, we saved and loaded plain text.

Today, we work with saved data that has structure: rows, columns, labels, and values that a program can select.

2



Three Tiny Data Shapes

Different formats, same simple information.

plain text 

Remember this task
Review later

Just words written line by line.

CSV 

task	note
Remember this task	high priority
Review later	no rush

Values separated by commas in rows and columns.

JSON 

```
{
  "task": "Remember this task",
  "note": "Review later"
}
```

Data in key-value pairs written with labels.

Saved Data has "Shape"

3



Save For Later



databases



custom JSON encoding



large data pipelines



advanced exceptions



configuration systems

Useful later, not today's required target.

Parked For Later

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Quick Review

When a program reads a file ...
the file becomes part of the program's world.

If the file is missing, damaged, or shaped differently than expected, the program needs a reasonable response.



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The File Is Part of the Program's World

```

graph LR
    A[Python program] --> B[data file]
    B -- missing --> C[program response]
    B -- wrong shape --> C
  
```

The program must respond clearly when the file is not what it expected.

Programming World Includes "Files"!

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What Counts as Success Today

Working baseline ->
compare structure ->
revise one meaningful thing ->
explain the improvement.

A useful revision should make the program:

clearer	easier to test	easier to change	easier to explain
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Reading Structured Data: A 5-Step Success Pattern

- 1 identify file shape
- 2 read the file
- 3 select useful values
- 4 handle one likely problem
- 5 explain what the program did

Common File Shapes

plain text Remember this task Review later	CSV task_name Remember this task, high priority Review later, no task	JSON { "task": "Remember this task", "priority": "Review later" }
--	--	---

The file shape affects the program's choices.

Today's Success

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What We Will Use Today

- ▶ CSV rows and columns
- ▶ JSON labels and values
- ▶ 'try' / 'except'
- ▶ selected-value output
- ▶ one clear error path

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Today's Structured Data Tools

Simple tools to read, use, and handle data with confidence.

BUILD SIMPLE, RELIABLE PROGRAMS

Use these tools together to build simple, reliable programs.

Toolbox

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What We Will "Skip" For Now

- ▶ databases
- ▶ custom JSON encoding
- ▶ large data pipelines
- ▶ advanced exception design
- ▶ configuration systems

Today we read small structured files and respond to likely problems.

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Save For Later

Useful later, not today's required target.

Parked For Later

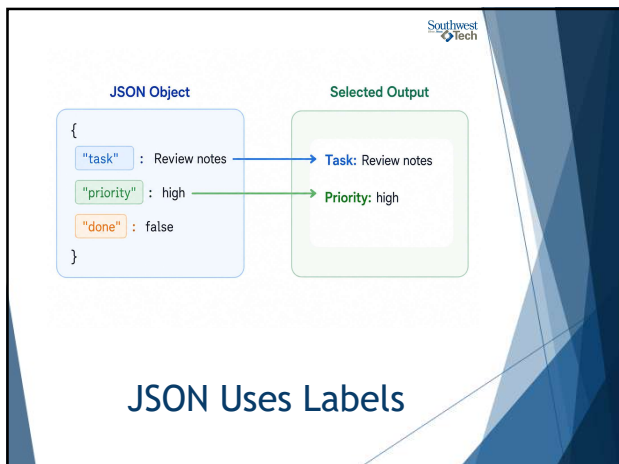
12



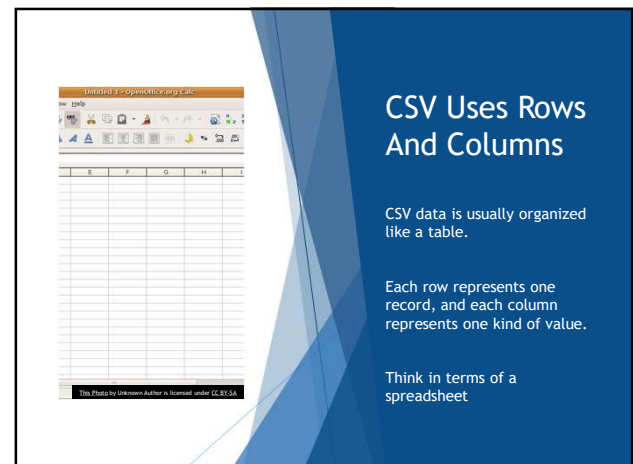
13



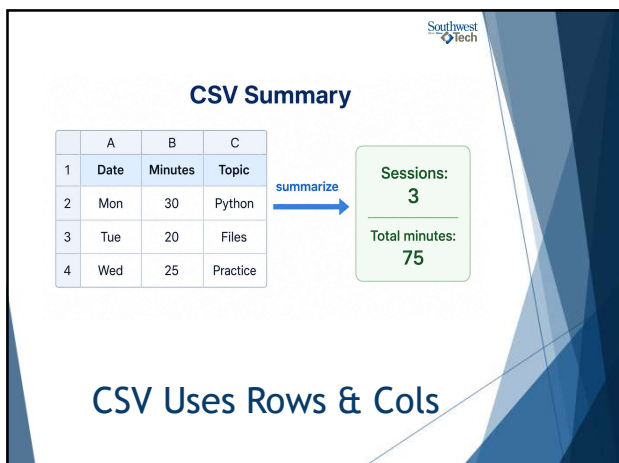
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Loaded Data Becomes Useful Output

Loaded data

```
id,task,done,order
1,study files,false,1
2,review notes,true,2
3,practice,true,3
... other fields ...
```

select /
summarize /
format

→

Useful output

Tasks completed: 3

Next task: Review notes

💡 Reading the file is only the first step.

"Loading" is NOT the Finished Result

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Demo #1

"Save And Load JSON Tasks"

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What to Watch For

- ▶ List of tasks
- ▶ Labels inside each task
- ▶ Selected values when loaded

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JSON Tasks: Save, Load, Select

JSON Tasks

```
task: "study files"
done: false
```

```
task: "review notes"
done: true
```

load JSON

→

Selected Output

2 tasks loaded

Completed: review notes

Demo 1: Save & Load JSON Tasks

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Demo #2

"Read A CSV Summary"

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What to Watch For

CSV data often supports small reports.

Instead of printing every row, the program can count, total, filter, or display selected records.

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CSV Summary

Date	Minutes	Topic
Mon	30	Python
Tue	20	Files
Wed	25	Practice

summarize →

Sessions: 3

Total minutes: 75

Demo 2: Read a CSV Summary

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Not Every Failure Means The Same Thing

Different problems need different explanations.

- ▶ *missing file*: the program cannot find what it was told to read
- ▶ *invalid data*: the file exists, but the contents do not match expectations

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Different Problems, Different Responses

File missing

Tell the user what file was not found

Invalid data

Tell the user the file contents need checking

Different problems need different, helpful messages.

Not Every Failure Means The Same Thing

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Demo #3

"Missing File And Invalid JSON"

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What to Watch For

Error handling should make the problem understandable.

The goal is not to hide failure. The goal is to respond clearly enough that a human knows what happened.

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Two Different File Problems

Missing file

File was not found.

Invalid JSON

File exists, but contents need checking.

Demo 3: Missing File & Invalid JSON

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Evidence Checklist

-  code file ☒
-  data file ☒
-  readable output ☒
-  data-shape explanation ☒
-  error-path note ☒
-  AI-use note, if used ☒

 Clear evidence shows what you built, what the data looked like, and how you handled it.

Evidence For A8 & A9

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Thank you! Have Fun!

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